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Micro-LARP

Micro-Lunar Autonomous Resource Prospector

Faux Fly Design and Test Report



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Senior Design Prototype Development

Abstract

The Faux Fly project provided an opportunity to develop, manufacture, and test a miniature rover before constructing the full senior design prototype. The Micro-LARP team developed a four-wheel-drive vehicle incorporating a 3D-printed chassis, battery power, solar panels, a scanning LiDAR sensor, wireless ground-station communication, and temperature-controlled ventilation. Preliminary calculations estimated motor current, electrical power, and forward speed. The team reported that the completed rover satisfied all ten demonstration requirements, including travel distance, slope climbing, operation on sand, and maneuverability. Integration and testing also identified limitations in motor startup behavior and wireless connectivity. These findings establish priorities for power-system characterization, communication reliability, and software development in the next phase of the LARP project.

I. Vehicle Description

A. Design Evolution

The initial concept used six wheels, battery power, four solar panels, a LiDAR scanner, a temperature sensor, and a drill. Four wheels were to be powered, with two unpowered wheels positioned between them. During manufacturing, the team removed the two unpowered wheels to reduce mass, mechanical complexity, and procurement time. The additional wheels would have required an axle and bearings that could not readily be obtained within the demonstration schedule. The drill was also omitted because of manufacturing and schedule constraints.

The completed vehicle used four powered wheels and a 3D-printed enclosure to support the mechanical and electrical components. Two solar panels were installed rather than four because of the available lid area. The panels were included in the prototype architecture; their contribution to operating power was not quantified in the reported tests.

B. Integrated Systems

Four DC gearmotors mounted along the sides of the chassis drove 3D-printed wheels. Traction tape was added to improve contact with the test surfaces. A stepper motor rotated the forward-mounted LiDAR sensor through a nominal sweep from -45° to 45°, with 0° defined as straight ahead. A small, 3D-printed orange fedora was added to the sensor assembly as a team design feature.

An internal temperature probe supplied measurements to a Raspberry Pi Pico 2 W. A rear-mounted fan provided active ventilation when the measured temperature reached approximately $25\,^\circ$C. Motor-driver boards controlled the wheel motors, and additional driver circuitry operated the scanning motor and fan. The Pico 2 W also supported wireless communication with a laptop ground station.

II. Preliminary Analytical Calculations

A. Motor Current and Electrical Power

The reported motor specifications give a rated-load current of $0.14\,$A per motor at $6\,$V [1]. For four motors, the estimated combined current at that operating condition is

$$I_{\text{motors}} = N_m I_m = 4(0.14) = 0.56\,\text{A}, \tag{1}$$

where N_m is the number of motors and I_m is the rated-load current per motor. Electrical input power is estimated using

$$P_m = V_m I_m = (6)(0.14) = 0.84\,\text{W}, \tag{2}$$

which gives a combined motor input power of

$$P_{\text{motors}} = N_m P_m = 4(0.84) = 3.36\,\text{W}. \tag{3}$$

The preliminary battery selection was reported as 3.7 V nominal, 4.2 V fully charged, with a maximum discharge current of 1.5 A. Although the calculated motor current is below that discharge rating, the motor and battery values refer to different voltages. Consequently, this comparison alone does not establish system compatibility. If a converter supplies the motor rail, battery-side current depends on its efficiency and the battery voltage:

$$I_b = \frac{P_{\text{motors}} + P_{\text{aux}}}{\eta V_b}, \tag{4}$$

where P_{aux} is the auxiliary electrical load, η is the converter efficiency, and V_b is the battery voltage. This relationship is included to clarify the power balance; auxiliary load and conversion efficiency were not measured during this project.

The preliminary estimate excluded LiDAR, control electronics, scanning-motor, and fan loads. Startup and stall currents were also not quantified. These omissions limit the calculation to a rated-load motor estimate rather than a complete power-system validation.

B. Forward Speed Estimate

Using a wheel radius $r = 0.05\,\text{m}$ and the reported motor speed of 60 rpm [1], the rotational frequency is

$$f = \frac{60}{60} = 1\,/\text{s}. \tag{5}$$

The corresponding ideal forward speed is

$$v = 2\pi r f = 2\pi(0.05)(1) \approx 0.314\,\text{m/s}. \tag{6}$$

This estimate assumes rolling without slip and operation at the specified rotational speed. Actual speed depends on supply voltage, loading, terrain, and wheel slip; no measured speed is provided in the report.

III. Manufacturing and Integration

A. Mechanical Fabrication

The team first developed a CAD model of the chassis and component arrangement, shown in Fig. 1. Structural components were fabricated using fused deposition modeling (FDM). Heat-set threaded inserts and metric hex screws secured the motors, fan, LiDAR assembly, and cover. Other electronics were temporarily secured inside the enclosure using masking tape. The lid incorporated openings for the scanning-motor assembly and solar-panel wiring.

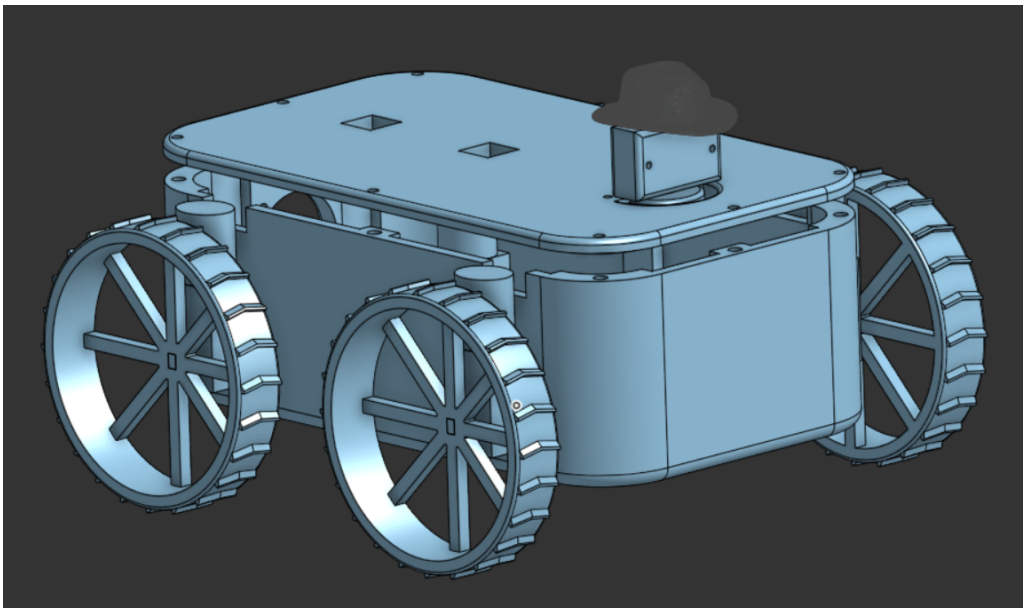
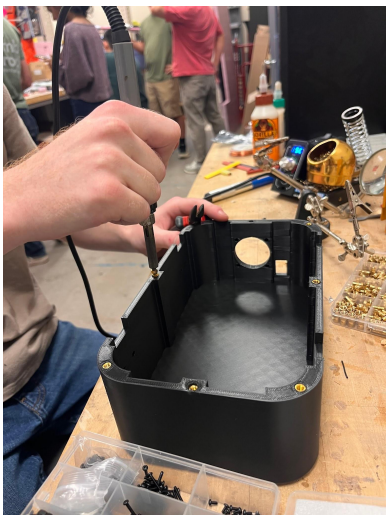


Figure 1: Final CAD model of the Micro-LARP rover.

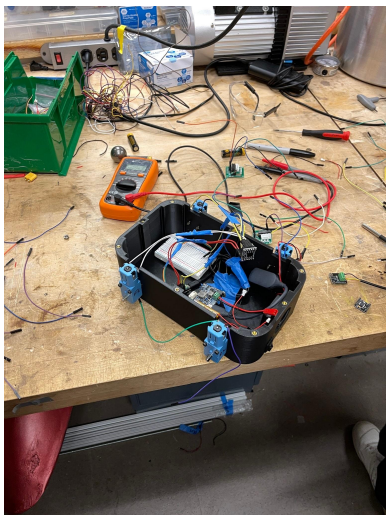
B. Electrical and Software Integration

The Raspberry Pi Pico 2 W served as the vehicle controller and communication interface. Peripheral connections were implemented using the Pico documentation [2]. Two motor-driver boards operated the wheel motors, with one assigned to the front wheels and one to the rear wheels. Additional driver modules operated the LiDAR scanning motor and switched the fan.

The temperature probe sent internal temperature measurements to the Pico, which transmitted telemetry to the laptop ground station. The fan was commanded on at 25 °C and off at 24.9 °C. The LiDAR range finder, described in the project records as having an 8 m range, also transmitted measurements through the Pico to the ground station. Team-written Python software provided vehicle control and telemetry handling.



(a) Heat-set insert installation.



(b) Component installation.



(c) Electrical integration testing.

Figure 2: Manufacturing and assembly activities.

IV. Team Member Contributions

Table 1 summarizes the responsibilities reported by the team.

Table 1: Team roles and contributions.

Area	Team members
CAD and manufacturing	Gabriel Flynn
Electrical and software	Bipin Dhakal and Nathaniel McIntire
Assembly and testing	Aamna Malik, Brayden Wilson, Darcy Johnston, and Floyd Henderson

V. Budget

The project budget was $200.00. The reported purchases totaled $199.20, leaving $0.80. Table 2 preserves the listed purchase amounts. Quantities, taxes, and shipping were not separately itemized in the project records.

Table 2: Reported parts expenditures.

Item	Cost (USD)
LiPo battery	9.99
Solar cell array	15.99
Waveshare solar power manager	20.99
Raspberry Pi Pico 2 W	18.95
DC motor controller	8.99
6 V DC gearmotor	16.99
LiDAR range finder	24.98
2,280-piece metric screw kit	19.99
520-piece metric threaded-insert kit	15.98
Boat rocker switch	6.99
40 mm, 5 V, 5,000 rpm fan	8.99
Temperature sensor	8.99
5 V DC stepper motor	7.99
20 AWG JST connector	6.40
1.25 mm JST connector	6.99
Total expenditure	**199.20**
Allocated budget	200.00
Remaining budget	**0.80**

VI. Safety Considerations

The principal electrical hazard was the lithium-polymer battery. Incorrect charging, short circuits, reversed polarity, or physical damage can cause overheating or fire. Relevant precautions include securing the battery against movement and damage, checking polarity before energizing the system, inspecting wiring for short circuits, and using appropriate battery charging and storage practices. A LiPo storage bag provides an additional precaution but does not eliminate battery hazards.

Component voltage and current ratings must also be respected. Motors, drivers, converters, and wiring should be evaluated against the actual supply voltage and anticipated operating and transient currents. The startup limitations

observed during testing reinforce the need to characterize the complete power path before increasing system loads.

VII. Test Results

The team reported successful completion of all ten Faux Fly demonstration requirements. Table 3 maps each requirement to the reported system or demonstration. The available test documentation consists of qualitative outcomes and photographs; measured distances, slope measurements, timing records, and repeated-trial statistics are not included.

Table 3: Reported demonstration outcomes.

No.	Requirement	Reported implementation or demonstration	Result
1	Battery power; hybrid source encouraged	LiPo battery; solar panels installed	Met
2	Ground-station communication	Pico 2 W wireless laptop interface	Met
3	Carried payload	Scanning LiDAR sensor	Met
4	Travel at least 10 ft	Four-wheel-drive locomotion	Met
5	Climb a 5% grade	Slope-climbing demonstration	Met
6	Travel on sand	Operation on simulated regolith	Met
7	Turn around a corner	Maneuvering demonstration	Met
8	Use a microcontroller	Raspberry Pi Pico 2 W	Met
9	Thermal control and survivability	Internal temperature probe and fan	Met
10	Team-written software	Custom Python control software	Met



(a) Driving on simulated Martian soil.



(b) Slope demonstration outside Engineering Building III.

Figure 3: Micro-LARP field demonstrations.

VIII. Challenges and Recommended Improvements

A. Motor Startup and Power Delivery

The wheels did not begin rotating at reduced motor commands but did move at full command. The team associated this behavior with startup loading and limitations in the power-delivery system. The Waveshare power manager was described as having a 3 A output rating; however, motor startup current and supply-rail voltage were not recorded. The observations therefore do not isolate a single cause. Available current, voltage drop, drivetrain friction, and motor-control settings should be evaluated in subsequent testing.

Future work should measure startup and loaded motor currents, monitor the supply voltage during acceleration, and include all auxiliary loads in the power budget. Battery, driver, and converter selection should then be based on those measurements. A higher-voltage battery with suitable voltage regulation may be considered after confirming component limits and the required transient current capability.

B. Wireless Communication

The team observed difficulty establishing reliable control when the laptop was more than approximately 15 ft from the rover before movement began. Connecting at shorter range generally supported continued operation unless the laptop disconnected from the rover network. These observations describe the tested configuration rather than a verified range limit of the Pico 2 W.

Enclosure placement and power-supply disturbances were identified as possible contributors, but neither was isolated experimentally. Follow-up tests should compare communication performance with the motors stopped and running, monitor the controller supply during motor startup, and evaluate controller placement within the enclosure.

IX. Conclusion

Micro-LARP provided practical experience in rover design, additive manufacturing, electrical integration, software development, and field testing. The completed prototype combined four-wheel-drive mobility, scanning LiDAR, wireless control, battery power, solar panels, and temperature-controlled ventilation within the reported $200 budget. The team reported satisfying all ten demonstration requirements.

The most useful findings concerned power delivery and communication reliability. Preliminary rated-load calculations did not fully describe startup demands or auxiliary loads, and wireless control performance depended on the tested operating conditions. Addressing these limitations through instrumented power tests and repeatable communication trials will support development of the full senior design prototype.

References

[1] Shenzhen Youmaike Technology Co., Ltd., *Gear Box Specification: TT Motor, 1:90 Gear Ratio*, Specification No. RJ00110608-R1, Oct. 28, 2017. URL: `https://cdn-shop.adafruit.com/product-files/3802/C11454_datasheet_TT1___90_metal.pdf` [accessed Oct. 27, 2026].

[2] Raspberry Pi Ltd., *Raspberry Pi Pico 2 W Datasheet*, Release 2, July 2026. URL: `https://pip-assets.raspberrypi.com/categories/1088-raspberry-pi-pico-2-w/documents/RP-008304-DS-3-pico-2-w-datasheet.pdf` [accessed Oct. 27, 2026].

A. Ground-Station Telemetry Sample

The following 50 records present an example of the telemetry exported to the ground station. A dash denotes an unavailable value, not a zero measurement. The angle column is an estimated scan angle. The fan column records the software command (1 = on, 0 = off), rather than a direct measurement of fan motion.

Table 4: Example telemetry imported from the Pico to the ground station.

Sample	Elapsed (s)	Temperature (°C)	Distance (cm)	Est. angle (°)	Fan command
1	1.427	–	–	–	1
2	2.429	21.62	–	–	0
3	3.429	21.62	35	35.3	0
4	4.429	21.62	42	36.8	0
5	5.429	21.62	–	–	0
6	6.43	21.62	–	–	0
7	7.432	21.62	52	-17.8	0
8	8.433	21.62	–	–	0
9	9.433	21.62	–	–	0
10	10.434	21.62	51	-18.8	0
11	11.435	21.62	28	-0.7	0
12	12.435	21.62	–	–	0
13	13.436	21.62	35	35.2	0
14	14.437	21.62	42	36.8	0
15	15.438	21.62	–	–	0
16	16.438	21.62	–	–	0
17	17.438	21.62	58	-16.9	0
18	18.439	21.62	–	–	0
19	19.44	21.62	–	–	0
20	20.44	21.62	51	-19.2	0
21	21.44	21.62	40	-1.2	0
22	22.44	21.69	–	–	0
23	23.441	21.69	34	34.8	0
24	24.442	21.62	41	37.6	0
25	25.442	21.62	–	–	0
26	26.442	21.62	–	–	0
27	27.443	21.62	59	-16.3	0
28	28.444	21.62	–	–	0
29	29.445	21.62	–	–	0
30	30.446	21.62	45	-23.3	0
31	31.446	21.62	63	-5.2	0
32	32.446	21.69	–	–	0
33	33.446	21.69	28	30.1	0
34	34.447	21.62	22	42.4	0
35	35.447	21.62	23	24.7	0
36	36.448	21.62	–	–	0
37	37.45	21.62	62	-8	0
38	38.451	21.62	37	-25.6	0
39	39.452	21.62	–	–	0
40	40.452	21.62	–	–	0
41	41.458	21.62	33	-14.9	0
42	42.459	21.69	–	–	0
43	43.46	21.69	–	–	0
44	44.462	21.62	34	34.7	0
45	45.462	21.62	37	38.2	0
46	46.463	21.69	21	22	0
47	47.465	21.69	–	–	0
48	48.466	21.69	35	-11.5	0
49	49.466	21.69	45	-28.5	0
50	50.466	21.69	–	–	0